

INTERNSHIP PRESENTATION

Java Software Development Projects

Presented By	Vanshika
Roll Number	25SCS2040002577
Programme	MCA (3rd Semester)
Batch	2025-2027
Organization	CodeAlpha
Internship Profile	Java Programming Intern

Virtual / Online Internship | 1 September 2026 – 30 September 2026

PRESENTATION OUTLINE

Contents



01

Internship Overview

02

Objectives of the Internship

03

Organization Profile

04

Technologies & Tools Used

05

Methodology

06

Projects Developed

07

Architecture / Design

08

Learning Outcomes

09

Conclusion & Future Scope

10

Offer Letter Proof

11

References

01 Internship Details

- Organization: CodeAlpha
- Profile: Java Programming Intern
- Mode: Completely online / virtual
- Duration: 1st September 2026 – 30th September 2026
- Offer Letter Issue Date: 31st August 2026

02 Student Details

- Name: Vanshika
- Roll No.: 25SCS2040002577
- Programme: MCA
- Batch: 2025–2027
- University: IILM University, Greater Noida, U.P.

- Enhance practical skills in Java programming and problem- solving.
- Apply OOP principles including classes, objects, encapsulation and methods.
- Organize application data using ArrayList and object- oriented structures.
- Build interactive user interfaces for real-time interaction.
- Implement input validation and meaningful system feedback.
- Perform calculations, manage data and present information clearly.
- Gain exposure to basic NLP techniques and rule- based chatbot logic.

CODE ALPHA

Internship Organization



Industry

- Software development & practical training



Focus

- Hands-on project development and programming practice



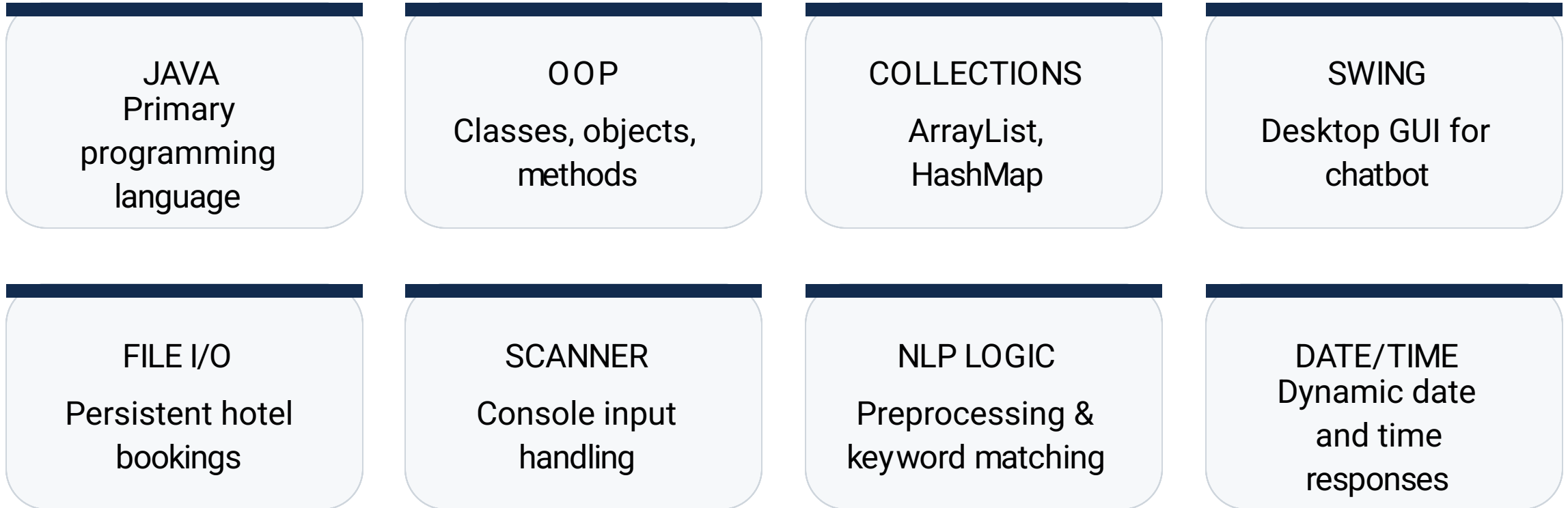
Mode

- Virtual / online internship

CodeAlpha served as the internship organization for the project work. The assignments were designed around practical software- development activities.

TECHNICAL STACK

Technologies & Tools Used





Incremental development was used: requirements → design → implementation → data handling → interface → testing → documentation.

01

Student Grade Tracker

Student data + grades + statistics

02

Stock Trading Platform

Market data + buy/sell + portfolio

03

Artificial Intelligence Chatbot

Java Swing + FAQ + rule-based logic

04

Hotel Reservation System

Search + booking + cancellation + File I/O

01 Key Features

- Console-based Java application.
- Student class with an ArrayList of grades.
- Accepts number of students and subjects.
- Validates grades within the 0–100 range.
- Calculates average, highest and lowest scores.
- Produces a formatted console summary.

→ Implementation Flow

- User Input
- Student Object Creation
- Grade Storage
- Grade Calculations
- Formatted Report

Core concepts: classes, objects, ArrayList, validation and statistical calculations.

02 Key Features

- Simulated trading environment.
- Stock objects store symbol, company name and price.
- User maintains cash balance, portfolio and transaction history.
- Buy checks available balance; sell checks holdings.
- Portfolio value can be calculated.
- Market price movement is simulated.

→ Implementation Flow

- Market Data
- User Action: BUY / SELL
- Portfolio & Balance Update
- Transaction Record
- Portfolio / Account View

Academic simulation only – the project models trading workflows rather than real financial trading.

03 Key Features

- Java Swing desktop application.
- FAQ knowledge base stored using HashMap.
- Lowercase conversion and punctuation removal.
- Extra spaces are normalized.
- Keyword similarity is used for intent matching.
- Date and time responses plus fallback messages.

→ Processing Flow

- User Message
- Text Preprocessing
- Keyword / Intent Matching
- Response Selection
- Response Displayed in GUI

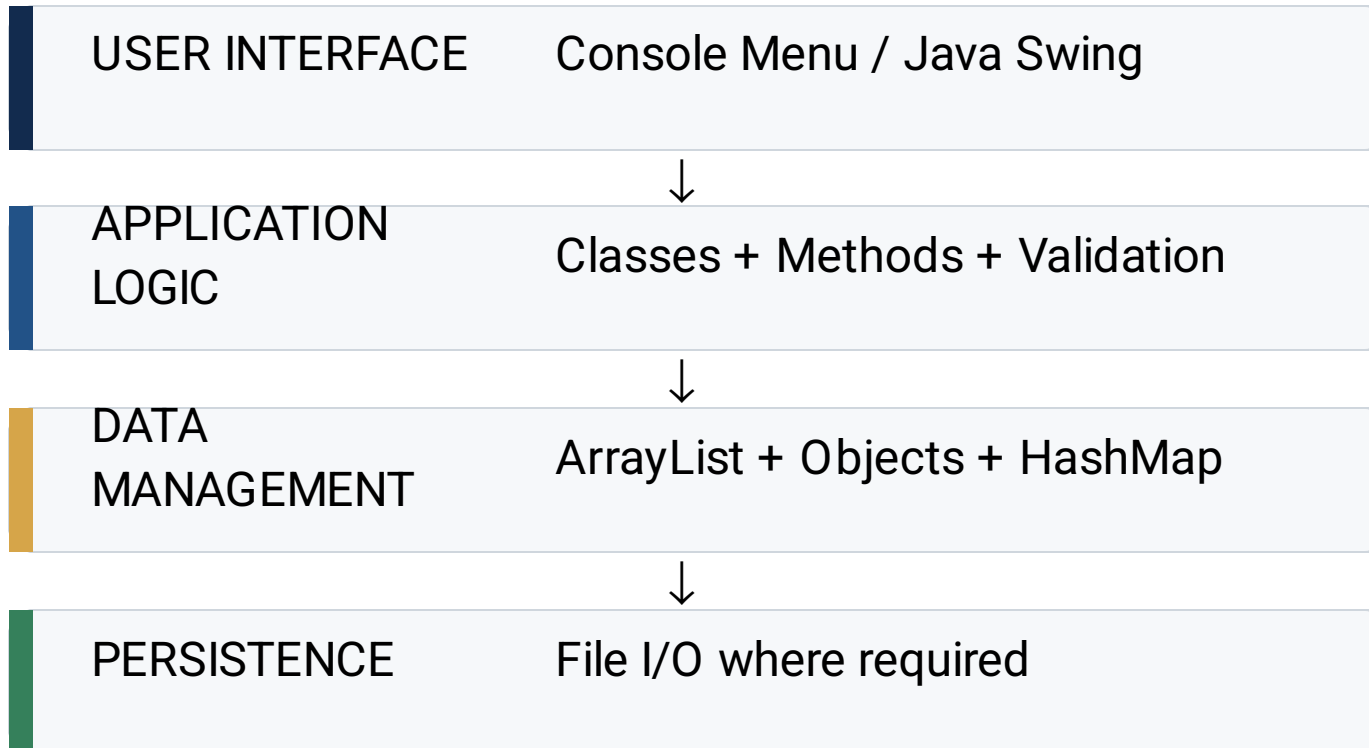
Important: the report describes this as rule-based / NLP-inspired logic, not a production machine-learning chatbot.

04 Key Features

- Console-based hotel reservation application.
- Room types: Standard, Deluxe and Suite.
- Searches available rooms by category.
- Creates a unique booking ID.
- Supports booking and cancellation.
- Simulates payment and stores active bookings.

→ Persistence Flow

- Load Saved Bookings
- Search Room Availability
- Create Reservation
- Simulate Payment
- Save to hotel_bookings.txt
- Reload on Application Start



The report uses modular, object-oriented design and selects data structures according to the nature of each application.



Technical Outcomes

- Improved Java syntax and OOP understanding.
- Practical experience with classes, objects, methods and collections.
- Console and GUI-based interaction.
- Input validation and application flow control.
- Basic NLP preprocessing and intent matching.
- File-based persistence.



Development Outcomes

- Translate problem statements into working solutions.
- Break applications into manageable modules.
- Test valid and invalid scenarios.
- Document project purpose, implementation and outcomes.
- Build confidence in structured software development.

CONCLUSION

- The four projects provide practical exposure to Java application development.
- OOP, data management, user interaction, validation and persistence were applied.
- Each project addresses a distinct real-world- inspired problem.

FUTURE SCOPE

- Add database integration and authentication.
- Improve GUI design and application usability.
- Integrate APIs and richer real-time data.
- Extend applications toward web-based deployment.



Offer letter details

- Name: Vanshika
- Profile: Java Programming Intern
- Mode: Virtual / Online
- Duration: 1 Sep 2026 – 30 Sep 2026
- Issue Date: 31 Aug 2026
- Student ID: CA/DF1/266225

Completion certificate: to be attached after internship completion

- Oracle Java Documentation — Java language and standard library concepts.
- Java Platform documentation — collections, Swing, file handling and date/time APIs.
- CodeAlpha internship task descriptions and project requirements.
- IILM University, School of Computer Science and Engineering — Internship Report Format (2026–27).

Note: This presentation follows the terminology and project details contained in the provided internship report and offer letter.

THANK YOU

Vanshika

MCA | Roll No. 25SCS2040002577

IILM University, Greater Noida, U.P.

CodeAlpha • Java Programming Internship